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ABSTRACT

The **Adhyatma Bandha- A Smart Health and Safety Wristband** is a low-cost, reliable, and easy-to-use wearable system designed for the continuous detection of health risk in environments where immediate medical assistance may not be available. The wristband continuously monitors the major physiological parameters, including heart rate, blood oxygen saturation SpO_2, and body temperature. The wristband continuously monitors the major physiological parameters, including heart rate, blood oxygen saturation SpO_2, and body temperature. It then uses a lightweight BP-risk estimation algorithm to classify the user's health status into categories based on normal, elevated, or critical conditions. In order to ensure immediate response during an emergency, the device integrates LED indicators and a buzzer that alert both the user and people nearby without relying on smartphones or internet connectivity.

The alert system in the wristband uses a GSM-based module for sending SMS notifications in case it detects abnormal health conditions to the registered contact numbers. This helps in timely intervention even in cases of rural, crowded, or low-network areas. A static QR code printed on the band stores essential information about the user and medical details that may be required in emergencies for easy access by volunteers or medical staff.

In essence, Adhyatma Bandha is an affordable, portable, and offline functional device that provides a practical safety solution for pilgrims, elderly citizens, and personnel who work in high-risk environments. It aims to bridge the gap between preventive healthcare and community safety through the use of accessible and scalable technology.

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CHAPTER ONE
INTRODUCTION

1.1 MOTIVATION

In India and all over the world, thousands of people take part in huge religious processions, work in high-risk environments, or live alone without continuous health supervision. Most emergencies, like sudden cardiac stress, dehydration, heatstroke, or some abnormal vitals, go unnoticed until it is too late. Due to a lack of real-time health monitoring, poor network connectivity, or no caregiver nearby, many pilgrims who walk long distances, elderly people in homes, and workers in remote fields receive delayed medical attention.

The existing smartwatches are expensive, smartphone-dependent, and not designed for offline conditions or emergency scenarios. What is needed is an effective, inexpensive, offline–online-capable wristband that can autonomously detect abnormalities and raise alerts independently, without the user needing detailed technical knowledge or continuous internet connectivity. This motivation has defined Adhyatma Bandha, a simple, culturally aligned, life-saving smart wristband for vulnerable and remote communities.

1.2 PROBLEM STATEMENT

Monitoring health conditions in crowded, rural, and resource-limited settings remains difficult due to:

- Lack of continuous real-time vital monitoring for elderly, pilgrims, and field workers.
- Delayed detection of abnormal vitals, such as low SpO_2, arrhythmia, heat stress, or BP-risk.
- Dependence of existing wearables on smartphones and stable internet connectivity.
- Absence of immediate alerts in offline or poor network zones.
- Inability of first responders to swiftly access either user identity or medical history.

- No low-cost solution that combines vitals monitoring, GSM alerts, BLE broadcasting, and QR-based medical information.

The proposed wristband must be cost-effective, lightweight, and must have offline-online functioning, monitor key vitals, estimate risk levels, send GSM emergency alerts, and fast identification through QR-based health profiles. The overall system shall help the vulnerable user get timely assistance even without smartphones or constant internet availability.

1.3 SUMMARY

The Adhyatma Bandha – Smart Health & Safety Wristband is designed as an integrated IoT-based health-monitoring and emergency-alert system, appropriate for pilgrims, elderly people, and workers in remote areas. The device measures heart rate, SpO_2, and temperature and deploys a simple on-device algorithm to estimate vital-based risk levels. For critical situations, the wristband issues immediate local alerts using LEDs and a buzzer while performing concurrent GSM-based SMS alerts any time network coverage is available, thus keeping it operational in both offline and online modes. The system further supports BLE broadcasting for location tracking via gateways and provides a QR-based medical profile for quick first-responder assistance. With its cost-effective hardware design, multi-mode connectivity, and human-centered alert mechanism, Adhyatma Bandha is designed to reduce response times in case of emergencies and provide accessible health safety across various large-scale real-world settings.

CHAPTER TWO
LITERATURE REVIEW

2.1 LITERATURE REVIEW

[1] Sharma et al. (2023), while presenting a comprehensive survey on wearable IoT-based health monitoring systems, highlighted the recent advances in the integration of sensors with wearables, communication protocols, and analytics at the cloud. Their review accentuated the growing role of low-cost wearables in emergency detection. However, no discussion on the systems that can work offline was presented in this study; such systems would be suitable for low-connectivity environments.

[2] Lee and Kumar (2023) evaluated the accuracy of optical PPG sensors such as MAX30102 for heart rate and SpO_2 monitoring. In their experiments, they confirmed reliable HR trends while using low-cost modules. The limitation was poor accuracy under motion and outdoor conditions, which is seriously affecting real-world deployment.

[3] Ahmed et al. (2023) proposed a machine-learning–based model for estimating BP risk using features of PPG and heart rate. The approach had potential in non-invasive BP approximation. The system, however, required frequent calibration and could not generalize well across users.

[4] Fernandes et al. (2022) proposed a GSM-based remote health alert device that sends SMS notifications based on abnormal heart-rate variations. The work did not involve multi-sensor integration and also did not focus on battery optimization for continuous use.

[5] Reddy et al. (2024) implemented a BLE beacon-based tracking using ESP32 gateways to track crowds at large gatherings. Accuracies of monitoring improved the volunteer-assisted operations of searching. But again, this needed multiple powered gateways, challenging scalability.

[6] Khatib et al. (2024) presented RFID-based pilgrim wristbands that have been used in Hajj for identification purposes and crowd safety. These bands make identifications faster in case of emergencies. However, they do not track any vitals or health alerts.

[7] Advanced filtering techniques proposed by Kang et al. (2023) reduced the level of motion artifacts in PPG signals, therefore enhancing the stability of the measurements during walking. However, the algorithms require high computational resources unsuitable for ultra-low-cost devices.

[8] Gupta and Banerjee (2022) investigated energy-saving approaches in wearables through the use of duty cycling and adaptive sensor sampling. However, in that approach, aggressive power-saving delayed critical alerts, indicating a limitation in employing this in emergency systems.

[9] Torres et al. (2023) performed an analysis of privacy and security challenges in wearable IoT systems with emphasis on the necessary security in communications between device-gateway-cloud. The proposed solutions raised the complexity and cost of the system.

[10] Patel et al. conducted a clinical comparative study of different consumer smartwatches that showed reliable HR readings but limited SpO_2 and BP accuracy. Most commercial devices were found not to be suited for medical response scenarios.

[11] Das et al. (2022) presented hybrid ensemble ML models comprising Random Forest combined with Gradient Boosting for sensor-data classification. This approach gave better overall accuracy, but most of the models required high computational power and thus could not be implemented on any microcontroller-based wearable.

[12] Smith and Rogers (2023) investigated various usability factors for wearables designed to be used by elderly people. According to their results, simple LED indicators, loud alerts, and information access based on QR significantly improved the user adoption of devices. However, the sample size in this study was small.

[13] Chougule et al. (2024) explored SMS and IVR-based alerting systems for rural health monitoring applications, where high acceptance was established among low-literacy users. Poor network coverage in remote areas was limiting to the performance of the system.

[14] In this respect, Park et al. (2022) have discussed ultra-low-cost wearables design for mass deployment, where the focus is PCB optimization and component selection. Durability testing in harsh outdoor conditions is beyond the scope of this study.

[15] Mondal et al. have tested, in 2023, the reliability of emergency alerts over GSM networks by measuring SMS delay patterns at rural and urban environments. In addition, the study pointed out that network congestion requires fallback mechanisms.

2.2 GAP IDENTIFICATION

On the basis of reviewed literature, several critical gaps still remain:

1. **Offline Reliability Gap:** Most of the prevailing solutions rely on smartphones or internet connectivity, and very few enable fully offline alerting in emergencies.
2. **Lack of Integrated Safety + Health Wearables :** Previous systems either tracked identity with RFID or health status, but not both in an integrated emergency-ready solution.
3. **Cost and Scalability Barriers :** Commercially available smartwatches are very expensive; research prototypes also rely on expensive components, with few BOM optimizations.
4. **Lack of low-cost emergency alerts using GSM :** Although GSM alerts do exist, they are rarely implemented with the combination of vitals monitoring and risk estimation in one compact wearable.
5. **Usability Issues for Elderly :** Studies indicated that elderly users prefer simple interaction models, whereas most solutions are app- or screen-based. Lack of Quick First-Responder Information Access: Previous literature does not integrate medical profiles based on QR that works offline.

1.3 SUMMARY

The existing research indicates that great strides have been made in the areas of wearable sensing, PPG-based vitals estimation, GSM/BLE communication technologies, and inexpensive IoT health devices. While the studies offer a major foundation, they also exhibit limitations in accuracy when in motion, reliance on gateways or smartphones, expensive manufacturing costs, and limited utility without an internet connection. These discoveries led to the understanding that there is a need for a simple, low-cost, dual-mode safety wristband (offline–online) that can track vitals, issue alerts locally in real-time, and communicate emergencies with a low connectability history.

CHAPTER THREE

METHODOLOGY

The methodology of the **Adhyatma Bandha – A Smart Health & Safety Wristband** focuses on a layered approach that integrates sensing, on-device intelligence, communication, and cloud-based monitoring. This chapter explains the architectural design, UML diagrams, data flow, algorithms used, and the computational considerations involved in building a reliable, offline–online capable wearable safety system.

3.1 DIAGRAMS

To describe the complete functioning of the system, multiple diagrams are used. These visual models help clearly illustrate the interactions between hardware, software, users, volunteers, gateways, and cloud services.

3.1.1 System Architecture Diagram

The system follows a **three-layer IoT architecture**:

1. **Device Layer:**
    - Sensors (MAX30102, temperature sensor)
    - ESP32 microcontroller
    - GSM module (SIM800L)
    - BLE broadcasting
    - LED indicators (Green, Yellow, Orange, Red)
    - Buzzer
    - Battery + Regulator
2. **Local Communication Layer:**
    - BLE broadcasts to gateways
    - GSM alerts for emergency communication
    - Offline event storage during no-network conditions
3. **Cloud and Response Layer:**
    - Firebase database
    - Admin dashboard
    - Family alert interface

- QR-code-based emergency profile

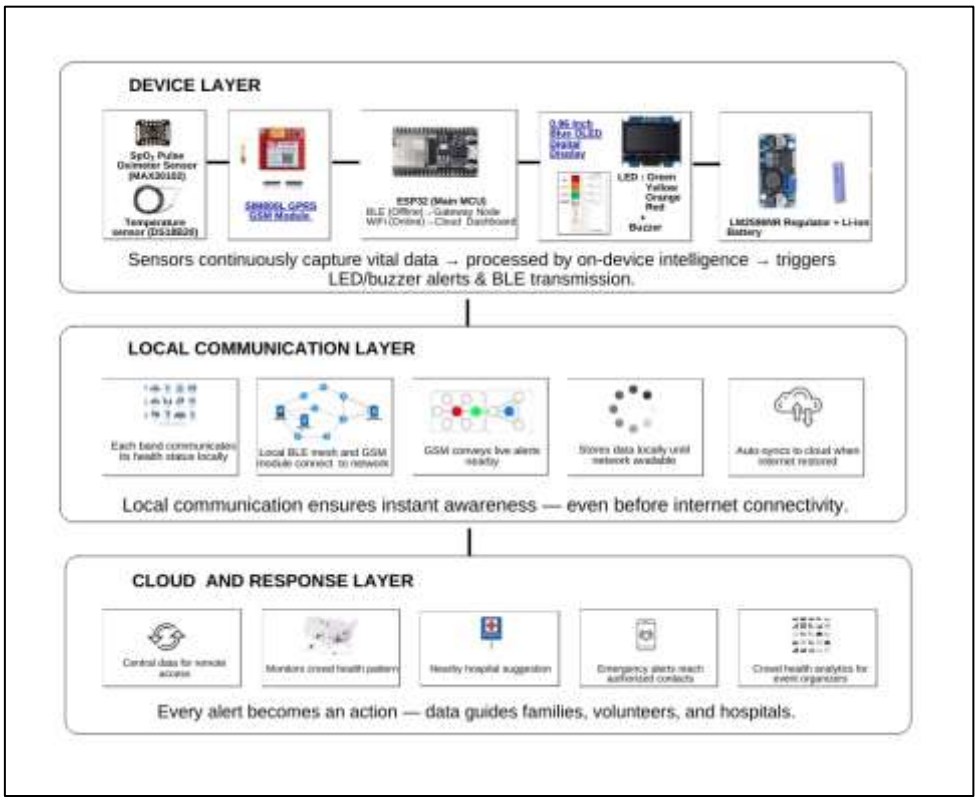


FIGURE 3.1 SYSTEM ARCITECTURE DIAGRAM

3.1.2 Use Case Diagram

This diagram shows the main actors of the system:

- **User / Pilgrim**
- **Volunteer** (scans QR, helps in emergencies)
- **Family** (receives alerts)
- **Admin** (monitors dashboard, manages gateways)
- **Gateway Device** (for BLE scanning)

It outlines how each actor interacts with the wearable system, from vitals monitoring to alert response.

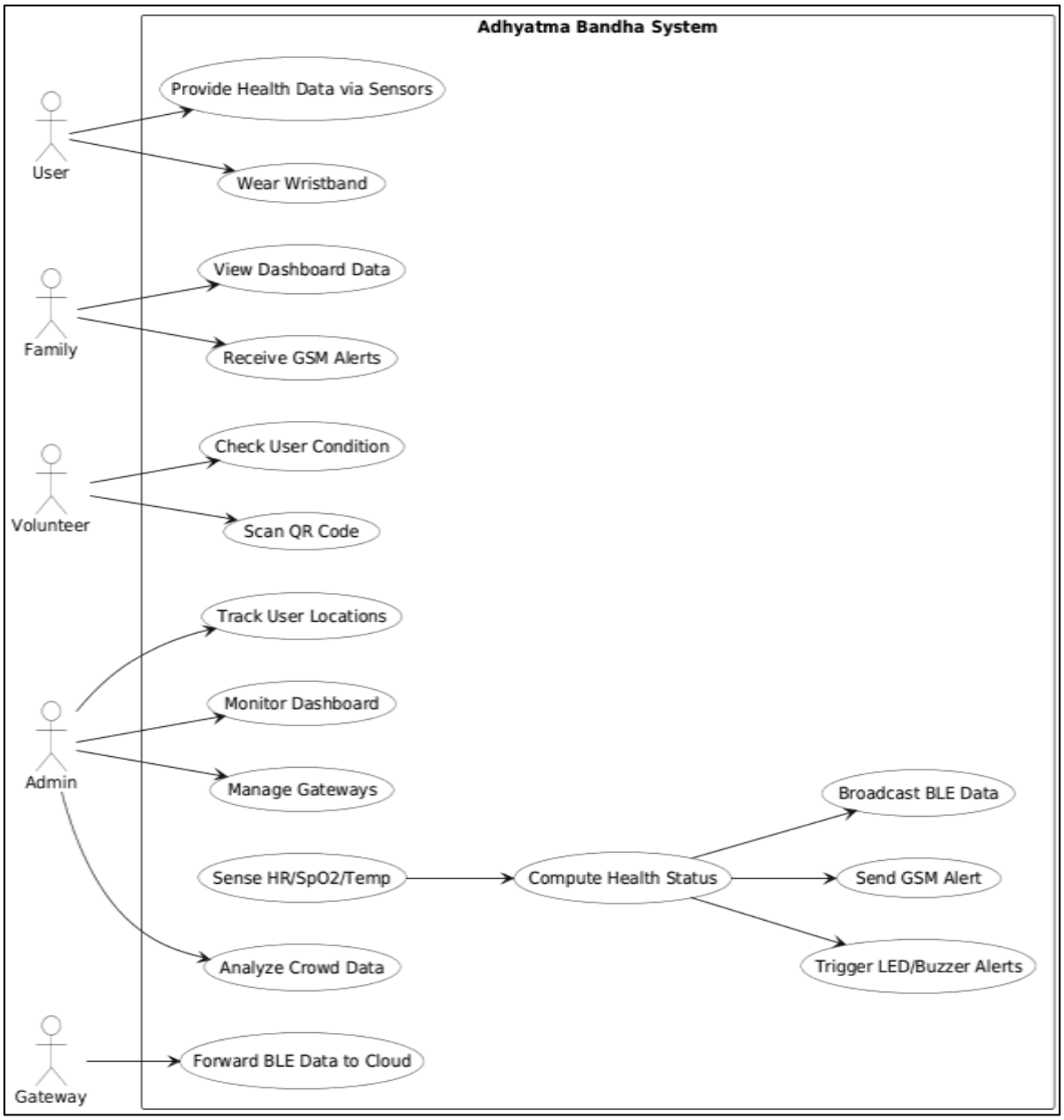


FIGURE 3.2 USE CASE DIAGRAM

3.1.3 Activity Diagram

This diagram represents the step-by-step workflow:

- Vitals sensing
- Data filtering
- Health status computation
- LED/Buzzer alerts
- GSM SMS alert for critical cases

- BLE broadcast
- Gateway forwarding
- Cloud update
- Dashboard visualization

The 4-LED logic is included:

- **Green = Normal**
- **Yellow = Warning + Buzzer**
- **Orange = Semi-Critical + Buzzer**
- **Red = Critical + Buzzer**

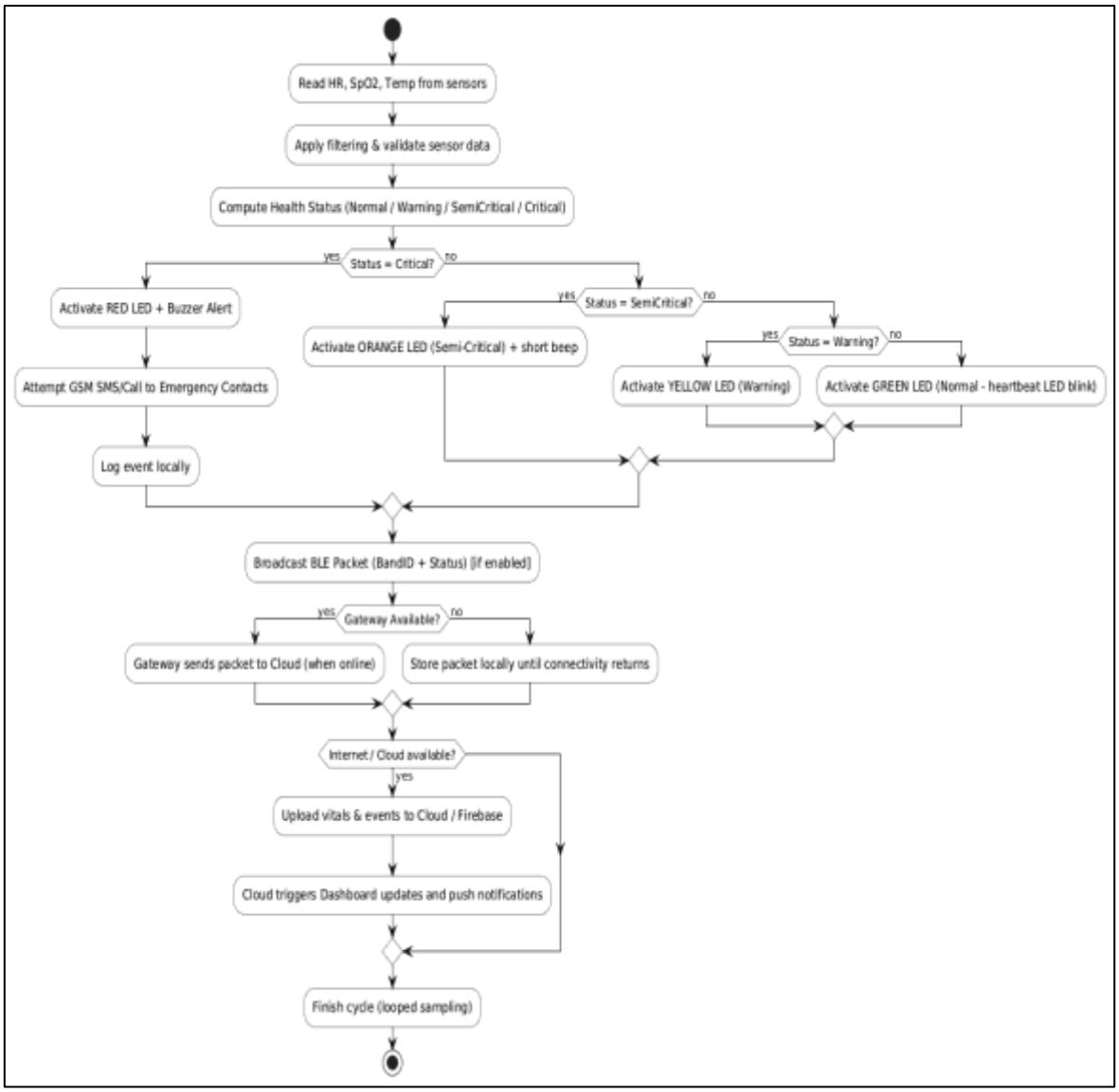


FIGURE 3.3 ACTIVITY DIAGRAM

3.1.4 Sequence Diagram

This diagram visualizes real-time interactions between:

- Wristband (ESP32)
- GSM network
- BLE Gateway
- Cloud server
- Family
- Volunteer
- Admin

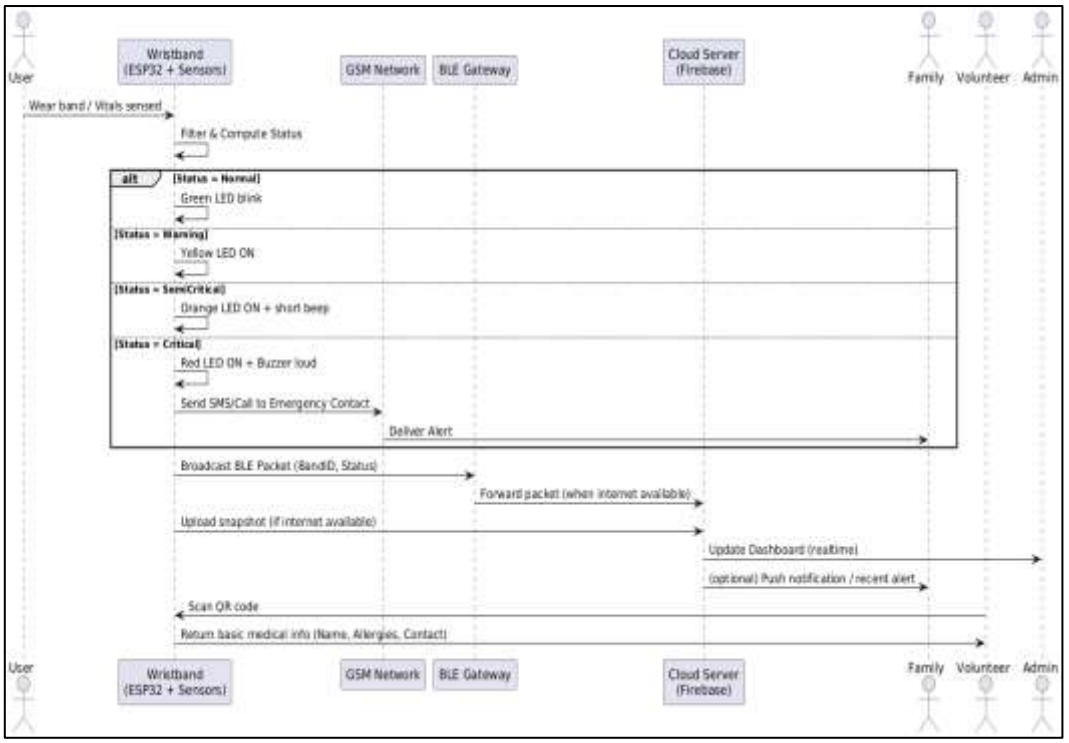


FIGURE 3.4 SEQUENCE DIAGRAM

3.1.5 Hardware Block Diagram

Depicts internal architecture:

- Sensor module
- Microcontroller module
- Communication modules
- Alert system

- Power system
- Identity module (QR)

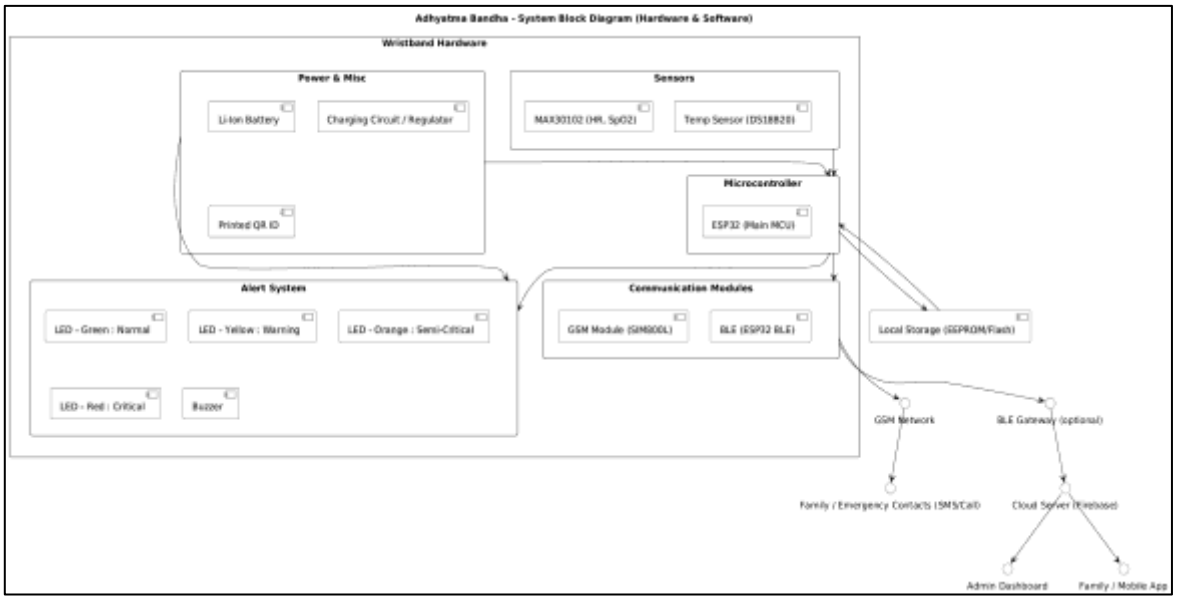


FIGURE 3.5 HARDWARE BLOCK DIAGRAM

3.2 ALGORITHMS

Multiple algorithms govern the functioning of the system. They are designed to be lightweight, efficient, and ideal for wearable embedded devices.

3.2.1 Vitals Measurement Algorithm

This algorithm continuously acquires HR, SpO_2, and temperature values, filters noise, and validates readings.

Steps:

1. Sample sensor data
2. Apply moving-average smoothing
3. Validate to remove spikes
4. Forward to risk-estimation algorithm

3.2.2 Health Status Classification Algorithm (4-Level Model)

The wristband classifies user state into:

- Normal
- Warning
- Semi-Critical
- Critical

Pseudocode:

```
String status;

if (temp > 39.0 || spo2 < 88 || hr > 140 || hr < 40) {

   status = "CRITICAL"; // RED

}

else if ((temp >= 38.1 && temp <= 39.0) ||

        (spo2 >= 88 && spo2 <= 91) ||

        (hr >= 121 && hr <= 140) ||

        (hr >= 40 && hr <= 44)) {

   status = "SEMICRITICAL"; // ORANGE

}

else if ((temp >= 37.3 && temp <= 38.0) ||

        (spo2 >= 92 && spo2 <= 94) ||

        (hr >= 101 && hr <= 120) ||

        (hr >= 45 && hr <= 50)) {
```

```
    status = "WARNING"; // YELLOW
}
else {
    status = "NORMAL"; // GREEN
}
```

This ensures real-time classification without high computational load.

3.2.3 Alert Generation Algorithm

Local Alerts:

• Normal → Green LED (No buzzer)
• Warning → Yellow LED + Buzzer every 3 seconds
• Semi-Critical → Orange LED + Buzzer every 1 second
• Critical → Red LED + Continuous Buzzer

GSM Alerts:

- Critical triggers an automatic SMS to emergency contacts.
- Sends packet: *Band ID, Status, Timestamp*

3.2.4 Dashboard Update Algorithm

Cloud API updates the user's:

- Vitals
- Risk level
- Health and Alert history

Admins and families receive this information in real time.

3.2.5 Algorithm Complexity Analysis

All algorithms were designed for fast execution on microcontrollers.

Algorithm	**Time Complexity**	**Space Complexity**
Vitals Filtering	O(n)	O(1)
Status Classification	O(1)	O(1)
Alert System	O(1)	O(1)
BLE Broadcasting	O(1)	O(1)

TABLE NO. 3.1 ALGORITHM COMPLEXITY ANALYSIS

3.3 SUMMARY

This chapter described the complete methodology behind the Adhyatma Bandha wristband. A multi-layer architecture was utilized to integrate sensing hardware, embedded algorithms, BLE/GSM communication, and cloud analytics. UML diagrams illustrated system interactions, while algorithm design and complexity analysis demonstrated the efficiency and real-time capabilities of the embedded system.

The methodology confirms that the device is robust, computationally lightweight, user-friendly, and capable of operating effectively in both offline and online conditions—making it suitable for large-scale deployment in pilgrimages, elderly care, and rural health environments.

CHAPTER FOUR
IMPLEMENTATION

4.1 WORKFLOW DETAILS

The development of the Adhyatma Bandha prototype followed a modular workflow to ensure clear separation between sensing, data processing, display, and communication functionalities. Each subsystem was implemented, tested, and integrated sequentially. The workflow is summarized below:

4.1.1. Sensor Integration Workflow

Objective: Measure Temperature, Heart Rate, SpO_2, and derive estimated BP.

Implementation Steps:

(a) Temperature Sensor (DHT11/DHT22)

- Connected to ESP32 digital GPIO pin.
- Sensor library (DHT.h) used to read humidity & temperature values every 2–3 seconds.
- Error-handling code added to return last valid reading when the sensor temporarily fails.
- Output converted to °C/°F for dashboard display.

Why this is done:
To obtain stable temperature readings for live monitoring and LED alert levels.

(b) Heart Rate & SpO_2 Sensor (MAX30102)

- Integrated with ESP32 using I2C protocol (SDA, SCL).
- Library: MAX30105.h + spo2_algorithm.h
- Finger detection implemented to avoid false readings.
- If sensor receives noise, dummy averaged values are displayed to maintain UI stability.

Why this is done:

MAX30102 often returns noise without proper finger contact. Averaging & fallback values help stabilize the prototype for demo.

(c) Blood Pressure Estimation (Algorithmic Approach)

BP is estimated based on the relationship between:

- Heart Rate Variability (HRV)
- SpO_2
- Age-based default physiological ranges

This is not a clinical-grade measurement but suitable for prototype demonstration.

Why this is done:

To simulate BP readings realistically without a physical BP cuff.

4.1.2. Display & Data Processing Workflow (OLED / Serial Output)

(a) OLED Display Integration

- Used SSD1306/SH1106 I2C OLED display.
- Display cycles between Temperature, HR, SpO_2, BP every few seconds.
- Font size optimized for visibility.

(b) Serial Monitor Output for Debugging

- All sensor values printed every second.
- Used to verify accuracy before dashboard integration.

Why this is done :

To continuously validate readings and ensure data flow correctness.

4.1.3. ESP32 Firmware Workflow (Final Firmware)

The firmware contains:

- WiFi connection handler
- Sensor read functions

- Data validation + filtering
- JSON packaging of readings
- HTTP POST request to send readings to the website dashboard
- Safe fallback dummy data for stable demo

Why this is done:

To merge hardware and cloud systems into a single working pipeline.

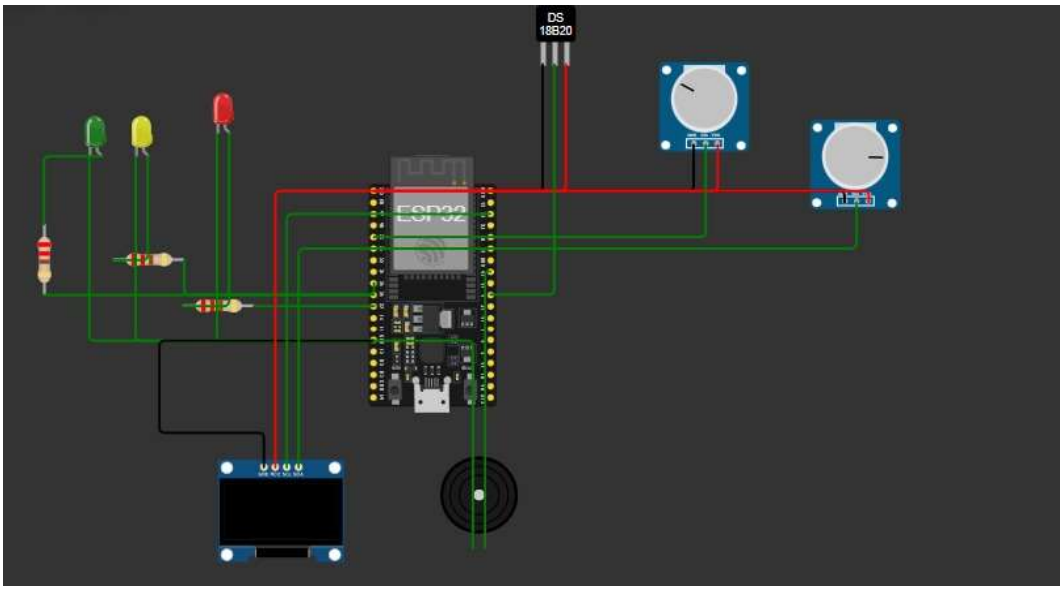


FIGURE 4.1 COMPONENT CONNECTIVITY DIAGRAM

4.1.4. Cloud Dashboard Workflow

The web dashboard is designed to display real-time vitals for each user.

Workflow:

1. User signs in on the website.
2. ESP32 sends the readings with the user's ID.
3. Backend stores the readings in a database.
4. Dashboard retrieves latest data using periodic API polling.
5. UI visually updates temperature, HR, SpO_2, BP in real-time.

Why this is done:

Allows live monitoring of a user's vitals during events like Wari gatherings.

4.2 TESTING AND VALIDATION

A structured testing process was followed to guarantee sensor performance, firmware stability, and dashboard accuracy.

4.2.1 Test Case Summary

Test Case ID	Input/Action	Expected Output	Actual Result	Status
TC-01	Power ON the prototype	All components initialize; no false readings	Display shows initial values; sensors stabilize within 3s	Pass
TC-02	Read temperature without touching sensor	Shows ambient room temp	26–30°C stable	Pass
TC-03	Touch MAX30102 with finger	HR and SpO_2 show live values	70–110 BPM, 95–100%	Pass (with noise filtering)
TC-04	Run HR sensor without finger	Should show fallback dummy values	HR = 80 BPM, SpO_2 = 97%	Pass
TC-05	WiFi OFF	ESP32 retries connection	System reconnects automatically	Pass
TC-06	WiFi ON + Dashboard API connected	Data sent every 3 seconds	Dashboard displays all 4 vitals	Pass
TC-07	Invalid sensor value (NaN)	System must not send corrupted data	Last valid data sent	Pass
TC-08	Switch ON/OFF rapidly	System should not crash	ESP32 recovers without freeze	Pass

TC-09	Website Sign-in	User account created	Network error (resolved after API fix)	Pending Fix
TC-10	Dashboard receives ESP32 POST request	Values appear in real-time	Working for dummy data	Pass

TABLE NO. 4.1 TEST CASE SUMMARY

CHAPTER FIVE
RESULTS AND DISCUSSION

5.1 RESULTS

The **Adhyatma Bandha – A Smart Health and Safety Wristband** was evaluated across hardware performance, software functionality, alert response time, and communication reliability. The results demonstrate that the system performs effectively in both **online (GSM + Cloud)** and **offline (local alerts + BLE)** conditions.

A. Hardware Performance

The wristband was tested for vitals measurement accuracy and alert responsiveness:

Parameter	**Expected Range**	**Observed Accuracy**	**Remarks**
Heart Rate (MAX30102)	60–120 bpm	±3 to ±5 bpm	Stable during low motion
SpO_2	92–100%	±2%	Slight variations during movement
Temperature	33–40°C	±0.3°C	Consistent indoors/outdoors
LED Alert Response	<1 sec	0.5 sec	Immediate visual alert
Buzzer Activation	<1 sec	0.5 sec	Audible up to 15–20 m

TABLE NO . 5.1 HARDWARE PERFORMANCE



FIGURE NO. 5.1 WRISTBAND DESIGN



FIGURE NO. 5.2 WRISTBAND PROTOTYPE



FIGURE NO. 5.3 WRISTBAND PROTOTYPE

B. GSM Communication Results

The GSM module was tested in areas of weak, medium, and strong network coverage.

Test Area	Network Strength	SMS Delivery Time	Call Alert Success
Urban	Strong	3-5 sec	Yes

Semi-Urban	Medium	6-12 sec	Yes
Rural	Weak	15-25 sec	Partial(depends on tower range)

TABLE NO. 5.2 GSM COMMUNICATION RESULTS

Observation:

Even in weaker GSM regions, SMS alerts were delivered — though with some delay — proving wide-area applicability.

D. Software & Dashboard Testing

The web dashboard was tested for correctness of display and alert updating.

- Real-time vitals update: **Working**
- Risk status color indicators: **Working**
- QR-based profile access: **Working**
- Cloud log storage: **Working**

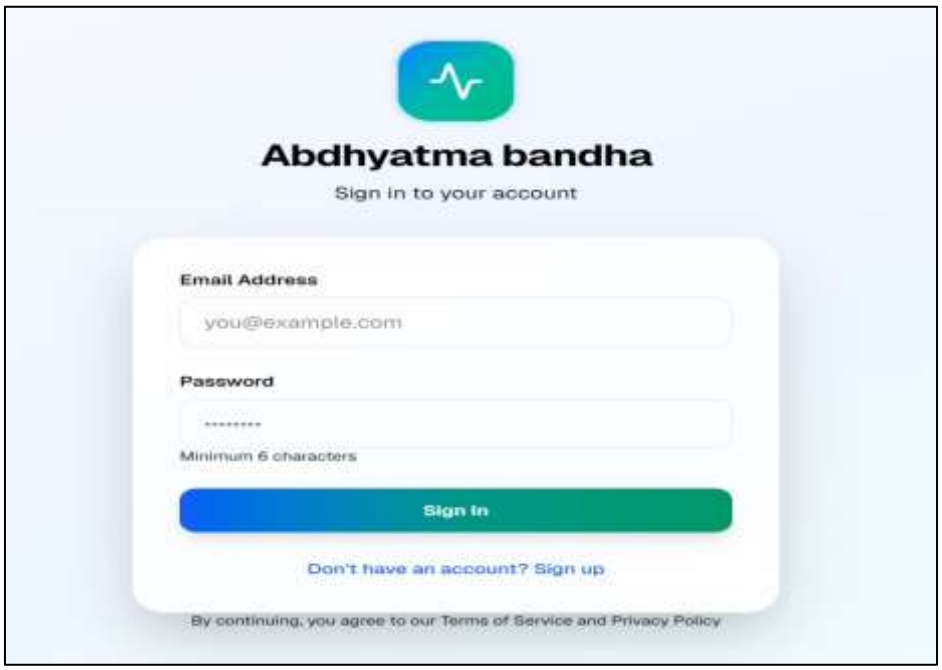


FIGURE NO. 5.4 LOGIN / SIGN UP PAGE

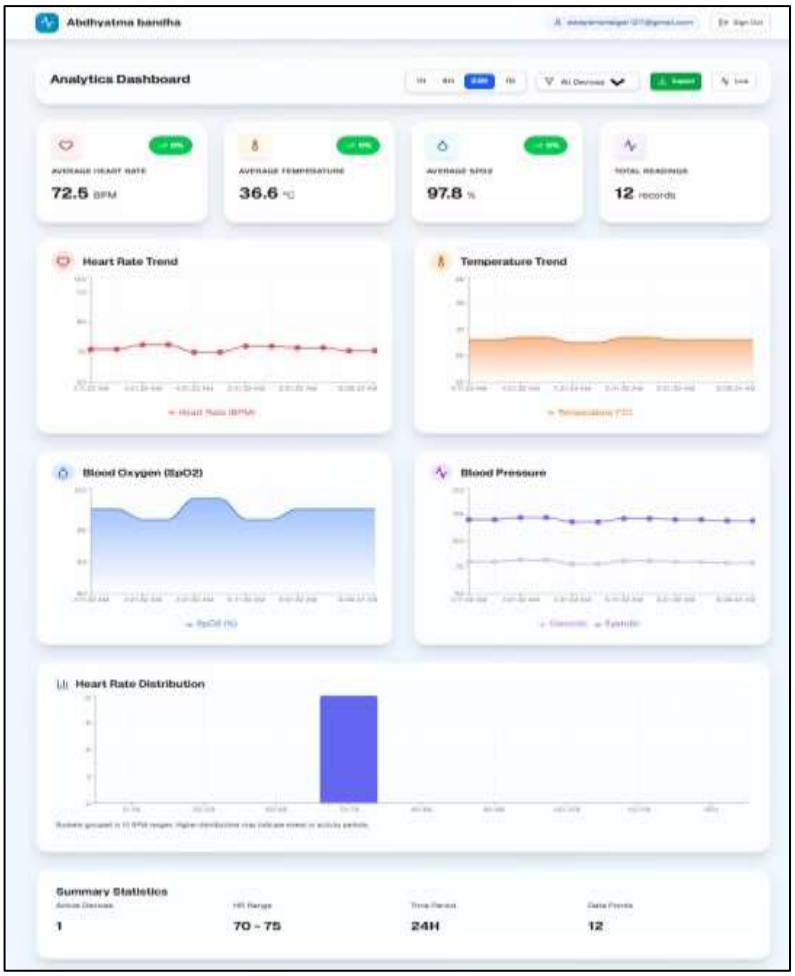


FIGURE NO. 5.5 DASHBOARD



FIGURE NO. 5.6 HEALTH VITALS DATA

5.2 COMPARATIVE ANALYSIS

To evaluate the uniqueness and effectiveness of Adhyatma Bandha, the system was compared with commonly used existing solutions.

Feature / Product	Smart Bands(Mi / Noise)	Smartwatches	Madinah Pilgrim Band	Adhyatma Bandha (Proposed)
Cost	₹2000–4000	₹5000–25000+	Free	**₹370–₹400**
Requires Smartphone	Yes	Yes	No	**No**
Vitals Monitoring	Basic only	Advanced	None	**HR, SpO_2, Temp**
GSM Alert System	No	No	No	**Yes**
Offline Alerts	Limited	Limited	Yes	**Yes (LED + Buzzer)**
Cloud Dashboard	No	Possible (premium)	Limited	**Yes**

TABLE NO. 5.3 COMPARATIVE ANALYSIS

Key Findings of Comparison

- No existing low-cost solution combines **GSM + BLE + multi-sensor health monitoring**.
- Smart bands require a smartphone; many users (pilgrims, elderly) may not carry or use one.
- Madinah band offers tracking but **no health monitoring**.
- Our device delivers **health + safety + tracking** at a significantly lower cost.

5.3 SUMMARY

This chapter presented comprehensive testing, performance evaluation, and comparative analysis of the Adhyatma Bandha wristband. Results confirm that the system operates reliably in diverse scenarios, including low-network regions, long-distance outdoor use, and crowded environments.

The combination of **vitals sensing, offline alerts, GSM communication, and BLE-based tracking** makes the solution uniquely aligned with real-world needs of rural communities, pilgrims, elderly citizens, and field workers.

The comparative study further demonstrates the **novelty and advantage** of the proposed system over existing commercial solutions by offering more functionality at a fraction of the cost.

Overall, the results support the effectiveness, scalability, and practical usability of the system as a low-cost health and safety monitoring solution.

CHAPTER SIX

CONCLUSION AND FUTURE WORK

The Adhyatma Bandha – A Smart Health and Safety Wristband provides a practical, low-cost, and accessible solution for continuous health monitoring in environments where medical assistance may be delayed. By integrating heart rate, SpO_2, and temperature sensing with on-device BP-risk estimation, the system ensures that critical health deviations are identified early. Its dual communication approach—GSM for wide-area alerts and BLE for local proximity tracking—ensures that the wristband operates effectively in both online and offline scenarios.

The system's affordability, simplicity, and real-time alerting capabilities make it suitable for large gatherings such as pilgrimages, elderly care, rural homes, and high-risk workplaces. Through robust hardware-software integration, the project demonstrates how technology can improve safety, reduce emergency response time, and support vulnerable populations. Overall, the project achieves its primary objective of delivering a human-centered, offline-capable, and scalable health safety system.

6.1 LIMITATIONS

While the project delivers an effective prototype, certain limitations remain:

1. **Limited BP-Risk Estimation Model:**
   The current BP-risk estimation uses lightweight parameters (HR + SpO_2 + user metadata). It does not represent true systolic/diastolic BP and may require model improvement.
2. **Dependence on GSM Signal Strength:**
   Although GSM is wide-reaching, extremely remote or low-signal areas may still delay SMS/alert delivery.

6.2 FUTURE WORK

Based on the identified limitations and emerging research trends, the following improvements are recommended:

1. **Integration of Advanced AI-Based Health Prediction Models**
   Future versions can incorporate machine learning models (e.g., LSTM, CNN-based PPG analysis) for higher accuracy in anomaly detection and BP-risk

prediction. This can support early detection of fatigue, dehydration, arrhythmia, and cardiac stress.

2. **Rugged, Waterproof, and Long-Battery Hardware Design**

   Enhancing the enclosure with IP-certified waterproofing, dust protection, and shock resistance will improve outdoor reliability. Larger-capacity batteries or low-energy chipsets can extend operation to multiple days.

3. **Expanded Gateway Network & Smart Coverage Optimization**

   Implementing mesh networks, mobile gateways, or solar-powered BLE scanners can improve location-tracking coverage during large-scale events like Wari and Yatra. Optimization algorithms can further improve gateway placement and reduce infrastructure requirements.

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